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Modeling Joint Health Effects of Environmental Exposure Mixtures with Bayesian Additive Regression Trees

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Abstract

Studying the association between mixtures of environmental exposures and health outcomes can be challenging due to issues such as correlation among the exposures and non-linearities or interactions in the exposure-response function. For this reason, one common strategy is to fit flexible nonparametric models to capture the true exposure-response surface. However, once such a model is fit, further decisions are required when it comes to summarizing the marginal and joint effects of the mixture on the outcome. In this work, we describe the use of soft Bayesian additive regression trees (BART) to estimate the exposure-risk surface describing the effect of mixtures of chemical air pollutants and temperature on asthma-related emergency department (ED) visits during the warm season in Atlanta, Georgia from 2011-2018. BART is chosen for its ability to handle large datasets and for its flexibility to be incorporated as a single component of a larger model. We then summarize the results using a strategy known as accumulated local effects to extract meaningful insights into the mixture effects on asthma-related morbidity. Notably, we observe negative associations between NO₂ and asthma ED visits and harmful associations between ozone and asthma ED visits, both of which are particularly strong on lower temperature days.

Keywords

Accumulated local effects; Air pollution; Asthma; Bayesian additive regression trees; Environmental mixtures

1 | INTRODUCTION

Asthma affected an estimated 25 million (7.7%) individuals in the United States in the year 2021 per the National Health Interview Survey.¹ In that same year, the Healthcare Cost and

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SOFTWARE

Software in the form of R code used to implement the proposed methods and produce the simulation results is available at <https://github.com/jacobenglert/softbart-mixtures-paper>.

FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

Utilization Project estimated a total of 5.8 million asthma-related emergency department (ED) visits, of which 1.4 million required hospitalization and 930,000 listed asthma as the primary diagnosis.² In this work we are interested in studying the marginal and joint associations between elevated concentrations of multiple airborne chemical pollutants on asthma-related ED visit rates in Atlanta, Georgia.

Previous studies of Atlanta and other U.S. cities have found harmful associations between asthma-related ED visits and environmental pollutants such as fine particulate matter of equal to or less than 2.5 μm in diameter ($\text{PM}_{2.5}$), nitrogen dioxide (NO_2), ozone (O_3), and carbon monoxide (CO), among others.^{3–9} The majority of these studies analyze exposures individually due to the challenges associated with modeling mixtures of correlated exposures. However, studies do often stratify analyses to study effect modification. For example, O'Lenick et al.⁸ reported that neighborhood-level socioeconomic status may affect the association between air pollution and pediatric asthma morbidity. Additionally, some studies have also reported modification of the effect of ozone on mortality by temperature.¹⁰

Modeling of environmental mixtures is often framed in one of two ways: targeting a restricted class of research questions using easily interpretable parametric models, or estimating the true exposure-response surface with fancier (but less interpretable) models based on Gaussian processes, regression tree ensembles, etc. The former includes summary index approaches such as quantile G-computation¹¹ and weighted quantile sum regression¹², while the latter encompasses tools like Bayesian kernel machine regression (BKMR)¹³, and more recently treed distributed lag mixture models (TDLMM)¹⁴ and multiple exposure distributed lag models.¹⁵ BKMR is most useful for estimating smooth exposure-response functions containing interactions and non-linearities, while the latter two focus primarily on interaction and lagged effects over discrete time intervals. In our review, existing methods typically address at most two of 1) non-linearity, 2) interaction, or 3) lagged effects (see Wilson et al.¹⁶ for an approach which seeks to address all three). We propose using soft Bayesian additive regression trees (BART)¹⁷ as an alternative to the BKMR approach for estimating interactions and non-linearities. While soft BART is computationally slower than traditional BART, the tree-based approach is more feasible than BKMR when working with datasets with a large number of observations, as in the motivating Atlanta dataset.

Following the fit of a mixture model, summarization of the estimated exposure-risk function with respect to one or two exposures often involves fixing the other exposures within the mixture to some quantile, or uses partial dependence statistics.¹⁸ These approaches tend to extrapolate to implausible mixture levels in the estimation process, particularly in the context of correlated exposures. We propose using accumulated local effects¹⁹ as an alternative approach that avoids this issue and has other benefits as well.

The main contribution of this work is to leverage a modeling approach based on Bayesian regression tree ensembles and subsequent summarization strategy for evaluating the effects of multi-pollutant mixtures on asthma morbidity in the city of Atlanta. We introduce the data for this application in Section 2 and outline the methodology in Section 3. We then

demonstrate the utility of the proposed approach through a simulation study (Section 4) before finally presenting our main findings from the application in Section 5.

2 | DATA

2.1 | Health Data

Patient-level billing records for ED visits to hospitals in the metropolitan Atlanta area from 2011-2018 were obtained from the Georgia Hospital Association. These data included admission date, billing address, International Classification of Disease (ICD) version 9 or 10 discharge diagnosis codes, and various patient characteristics. We restricted the ED visit data to only include visits containing an asthma diagnosis (ICD-9 code 493; ICD-10 code J45) and occurring during Atlanta's warm season (April-October). These visits were then aggregated by ZIP code and date for the analysis. These data have previously been used for various asthma and air pollution association studies.^{3,7-9,20}

2.2 | Air Pollution Data

Air pollution concentration data are collected daily and include fine particulate matter with diameter 2.5 μm and smaller ($\text{PM}_{2.5}$, 24-hr average, $\mu\text{m}/\text{m}^3$), ozone (O_3 , 8-hr max, ppm), nitrogen dioxide (NO_2 , 1-hr max, ppb), and carbon monoxide (CO , 1-hr max, ppb). The estimates are derived from the data fusion model described by Senthilkumar et al.²¹, which utilized simulations from the Community Multiscale Air Quality Model monitoring data from the Environmental Protection Agency's Air Quality System database. The data product is available at a 12km gridded spatial resolution and is linked to each ZIP code based on area-weighted averaging.

2.3 | Other Data

Maximum daily temperature is obtained from Daymet.²² The 1km gridded product was spatially averaged within each ZIP code, and linked to the ED visit data by both date and ZIP code. Annual ZIP code-level estimates of total population and the percent of the population below the poverty level are obtained from the 5-year American Community Survey for years 2011-2018.

3 | METHODS

3.1 | Soft BART

Bayesian additive regression trees (BART) is a nonparametric machine learning approach which approximates complex functions using sums of shallow Bayesian decision trees.²³ In this sense, the approach is similar to *boosting* from the machine learning literature. In recent years BART has soared in popularity due to its performance on prediction, classification, and causal inference tasks.²⁴ Ultimately, BART is a tree-based approach that can only approximate smooth functions with rigid fits. Linero and Yang¹⁷ propose a *soft* version of BART, which adapts to smooth functions better than traditional BART by swapping traditional decision trees for soft decision trees.²⁵ In a soft decision tree, the prediction for an observation is a weighted average of all of the leaf node parameters, where the weights are defined as the probability an observation is mapped to each leaf node as determined by,

say, a logistic gating function.¹⁷ When compared to the deterministic predictions obtained from a traditional decision tree, this has the effect of smoothing over the otherwise rigid decision rules that form the binary tree. Since we generally expect the exposure-risk surface to be smooth, we opt to use this version of BART in our implementation.

3.2 | Negative Binomial Regression with BART

We propose using the following Poisson regression model with gamma-distributed mean:

$$\begin{aligned} Y_{ij} | \theta_{ij} &\sim \text{Poisson}(\theta_{ij}) \\ \theta_{ij} &\sim \text{Gamma}(\xi, \exp(\eta_{ij})) \\ \eta_{ij} &= \log(\text{Pop}_{ij}) + \mathbf{X}_{ij}^T \boldsymbol{\beta} + f(\mathbf{Z}_{ij}) + v_i, \end{aligned} \quad (1)$$

where Y_{ij} and θ_{ij} represent the observed and expected asthma ED counts in region i on day j , respectively, for $i = 1, \dots, I$ and $j = 1, \dots, J$. It follows that marginally, Y_{ij} is distributed as a negative binomial random variable with parameters ξ and $p_{ij} = \exp(\eta_{ij}) / [1 + \exp(\eta_{ij})]$. The log expected counts are offset by $\log(\text{Pop}_{ij})$, the population of region i at time j , and the overdispersion in the counts is represented by ξ . Potential confounders such as federal holidays and socioeconomic factors are represented by the $p \times 1$ vector $\mathbf{X}_{ij}$, while all exposures are represented by $q \times 1$ vector $\mathbf{Z}_{ij}$. To account for additional unexplained variation in the counts due to location we include a ZIP code specific random intercept v_i .

The confounders are modeled linearly with regression coefficients given by the $p \times 1$ vector $\boldsymbol{\beta}$. The exposures are modeled using soft BART¹⁷. Specifically, in Equation (1) we use $f(\mathbf{Z}_{ij}) = \sum_{t=1}^T g(\mathbf{Z}_{ij}; \mathcal{T}_t, \mathcal{M}_t)$. The $\{\mathcal{T}_t, \mathcal{M}_t\}_{t=1}^T$ parameters correspond to the tree structures and scalar-valued leaf node parameters associated with the soft BART model, and g is the function which maps a set of exposures $\mathbf{Z}_{ij}$ to its prediction from a single soft decision tree. Our approach is similar to that proposed by Mutiso et al.²⁶, with the main difference being that we substitute the BKMR, which uses Gaussian processes, for soft BART.

3.3 | Model Estimation

The model described in Equation (1) is estimated with a Markov chain Monte-Carlo (MCMC) algorithm consisting of a mix of Gibbs and Metropolis-Hastings (MH) steps. BART (and by extension, soft BART), has been extended to many different outcome types since the original BART model was proposed. However, these extensions typically require conditional conjugacy between the outcome distribution and prior distribution on the individual leaf node parameters to facilitate the sampling of tree structures. Since the negative binomial likelihood does not itself admit a conditionally conjugate prior, we adopt the framework proposed by Pillow and Scott.²⁷ We augment the observed outcome y_{ij} with latent weights ω_{ij} sampled from a Pólya-gamma (PG) distribution.²⁸ Specifically, if we independently draw $\omega_{ij} \sim \text{PG}(y_{ij} + \xi, \eta_{ij})$, then the latent outcome $y_{ij}^* = \frac{y_{ij} - \xi}{2\omega_{ij}}$ follows a normal distribution with mean η_{ij} and variance $1/\omega_{ij}$. This leads to a convenient Gibbs sampler based on the Bayesian backfitting approach of Hastie and Tibshirani²⁹ for updating $\boldsymbol{\beta}$, $\{\mathcal{M}_t\}_{t=1}^T$, and

ν , when multivariate normal prior distributions are chosen. The full forms of these conjugate updates are provided in Appendix A of the Supplementary Material.

In BART, new tree structures are sampled from their marginal distribution and updated using an MH step by first integrating out the leaf node parameters. The tree structures themselves are proposed from a so-called *branching process* prior, which grow, prune, or perturb the existing structures from the previous iteration.^{23,30} A detailed explanation of how this approach works with weights, such as those introduced by the Pólya-gamma data augmentation scheme, is outlined in Bleich and Kapelner.³¹ The tree prior used for soft BART is more involved, optionally including hyperparameters and hyperpriors responsible for the degree of smoothness and/or sparsity, but the general concept is the same. One of the benefits of BART is that default priors tend to work well in a variety of circumstances. We use default priors for updating $\{\mathcal{T}_n, \mathcal{M}_i\}_{i=1}^T$ as suggested by Linero and Yang.¹⁷

We assign a mean-zero proper conditional autoregressive (pCAR) prior for the ZIP code-level random intercepts $\nu = (\nu_1, \dots, \nu_I)^T$. In mathematical terms, $\nu \sim \text{MVN}(\mathbf{0}, \Sigma_\nu)$ with $\Sigma_\nu = \tau^2(\mathbf{D} - \rho\mathbf{W})^{-1}$, where $\mathbf{W}$ is the $I \times I$ first-order adjacency matrix, and $\mathbf{D}$ is the $I \times I$ diagonal matrix containing the number of neighbors for each region. This allows for a portion of the variability in the response unexplained by the predictors to be attributed to unmeasured spatial factors. We assign an inverse Gamma and discrete uniform hyperprior to τ^2 and ρ , respectively.

Lastly, we update the dispersion parameter ξ using the conjugate sampling routine described by Zhou et al.³² This technique relies on expressing the $Y_{ij} \sim \text{NB}(\xi, p_{ij})$ marginal distribution as a compound Poisson distribution. The details for this step and the others described in this section are outlined in Appendix A of the Supplementary Material. A summary of a single MCMC iteration is provided in Algorithm 1.

3.4 | Model Interpretation with Accumulated Local Effects

When using flexible methods which target the response-surface directly, interpretation of the resulting fit requires just as much thought as the estimation itself. In the environmental mixtures setting, it is common to focus on the marginal effect of a single exposure on the outcome by evaluating the exposure-response function at several levels of the chosen exposure and plotting the result. This strategy can also be used for studying the joint effects of two exposures using, say, contour plots. Analyzing/visualizing joint effects of more than two continuous exposures is rather difficult, and thus is not as common. Mathematically, to evaluate the effect of a set of exposures $\mathbf{Z}_k$ on an exposure-response function f , where k indexes the subset of the q exposures of interest, we compute $\hat{f}(\mathbf{z}_k | \mathbf{Z}_{[-k]})$ for several values $\mathbf{z}_k$ in the observed multivariate range of $\mathbf{Z}_k$.

Algorithm 1

One MCMC iteration of the Soft BART negative binomial model

1: **Input:** $\mathcal{D} = \{\mathbf{y}, \mathbf{X}, \mathbf{Z}, \text{Population}, \mathbf{D}, \mathbf{W}\}, \beta, \nu, \tau^2, \rho, \{\mathcal{T}_i, \mathcal{M}_i\}_{i=1}^T, \xi, \alpha_\tau, \beta_\tau, \alpha_\xi, \beta_\xi$

2: **for** $i = 1, \dots, I$ and $j = 1, \dots, J$ **do**
 3: Draw $\omega_{ij} \sim \text{PG}(y_{ij} + \xi, \eta_{ij})$.
 4: Compute $y_{ij}^* = \frac{y_{ij} - \xi}{2\omega_{ij}}$.
 5: **end for**
 6: Form $\mathbf{\Omega} = \text{diag}(\omega_{11}, \dots, \omega_{1J}, \dots, \omega_{I1}, \dots, \omega_{IJ})$.
 7: Draw $\boldsymbol{\beta} \sim \text{MVN}(\boldsymbol{\mu}_\beta^*, \boldsymbol{\Sigma}_\beta^*)$, where $\boldsymbol{\mu}_\beta^*$ and $\boldsymbol{\Sigma}_\beta^*$ are defined as in Appendix A.3.
 8: Draw $\mathbf{v} \sim \text{MVN}(\boldsymbol{\mu}_v^*, \boldsymbol{\Sigma}_v^*)$, where $\boldsymbol{\mu}_v^*$ and $\boldsymbol{\Sigma}_v^*$ are defined as in Appendix A.4.
 9: Draw $\tau^2 \sim \text{IG}(\alpha_\tau + I/2, \beta_\tau + \mathbf{v}^T(\mathbf{D} - \rho\mathbf{W})\mathbf{v})/2$
 10: Draw ρ from its discrete posterior distribution described in Appendix A.6.
 11: **for** $t = 1, \dots, T$ **do**
 12: Propose/update $\mathcal{T}_t$, $\mathcal{M}_t$, and any associated hyperparameters governing the degree of smoothness and/or sparsity as described by Linero and Yang¹⁷.
 13: **end for**
 14: Draw $L_{ij} \sim \text{CRT}(\xi, y_{ij})$ for $i = 1, \dots, I, j = 1, \dots, J$.
 15: Draw $\xi \sim \text{Gamma}(\alpha_\xi + \sum_{i=1}^I \sum_{j=1}^J L_{ij}, \beta_\xi - \sum_{i=1}^I \sum_{j=1}^J \ln(1 - p_{ij}))$.

In settings where there are more than two exposures, a decision must be made regarding the treatment of the “other” exposures $\mathbf{Z}_{[-k]}$ when evaluating the exposure-response function for 1-2 exposures of interest $\mathbf{Z}_k$. A common choice is to set $\mathbf{Z}_{[-k]}$ to some fixed value (e.g., their observed medians) while varying $\mathbf{Z}_k$. In fact, one might set $\mathbf{Z}_{[-k]}$ to multiple values (e.g., median and 95th percentile), and plot $\hat{f}(\mathbf{z}_k | \mathbf{Z}_{[-k]})$ for each setting. This approach is not ideal since the exposure-response function ultimately depends on the selected values for $\mathbf{Z}_{[-k]}$, of which there are many choices for each exposure in the model - none of which are the perfect choice, and many of which are poor choices.

Another option is to calculate partial dependence (PD) functions.¹⁸ PD functions target the average exposure effect across the marginal distribution of the observed data, setting $\mathbf{Z}_{[-k]}$ to their observed values $\mathbf{Z}_{ij, [-k]}$. Thus, PD functions do not require the user to choose the values of $\mathbf{Z}_{[-k]}$, and the resulting estimates naturally incorporate the variability in $\mathbf{Z}_{[-k]}$ across specified values of $\mathbf{Z}_k$. One of the major limitations of PD functions is that they are computationally burdensome, requiring predictions for every observation in the study at every value considered for $\mathbf{Z}_k$.

Both the fixed-value and PD approaches run the risk of extrapolating when evaluating $\hat{f}(\mathbf{z}_k | \mathbf{Z}_{[-k]})$, particularly if the exposures are correlated (which they often are). By this we mean that some of the evaluations of $\hat{f}(\mathbf{z}_k | \mathbf{Z}_{[-k]})$ are made on implausible exposure profiles. This is a general issue for assessing covariate effects in black-box supervised learning models. One approach that has been proposed to combat this issue is accumulated local effects (ALE).¹⁹ The estimands for the partial effect of a single exposure Z_k at some level z_k for each of the three approaches are provided in Equations (2), (3), and (4).

$$f_{k, Fixed}(z_k) \equiv f(z_k, \mathbf{Z}_{[-k]}) \quad (2)$$

$$f_{k, PD}(z_k) \equiv \mathbb{E}[f(z_k, \mathbf{Z}_{[-k]})] \quad (3)$$

$$f_{k, ALE}(z_k) \equiv \int_{z_{min,k}}^{z_k} \mathbb{E}\left[\frac{\partial f}{\partial Z_k}(Z_k, \mathbf{Z}_{[-k]}) | Z_k = z'_k\right] dz'_k \quad (4)$$

For estimation purposes, Equation (2) only requires on evaluation of $\hat{f}$ at some fixed value $(z_k, \mathbf{z}_{[-k]})$, Equation (3) requires an evaluation of $\hat{f}$ at $(z_k, \mathbf{Z}_{ij, [-k]})$ for all i, j , and Equation (4) requires an evaluation of $\hat{f}$ at $(z_k^{lower}, \mathbf{Z}_{ij, [-k]})$ and $(z_k^{upper}, \mathbf{Z}_{ij, [-k]})$ for all i, j s.t.

$Z_{ij,k} \in (z_k^{lower}, z_k^{upper})$. The latter is a result of approximating $\frac{\partial f}{\partial Z_k}$ with small finite differences.

For more information regarding ALE computation, we refer the reader to Apley and Zhu.¹⁹ When using a Bayesian approach, each of Equations (2), (3), and (4) would be evaluated for each sample from the posterior distribution. This allows one to obtain point-wise posterior means and uncertainty estimates at each z_k .

Extending the fixed-value and PD approaches to two or more exposures is straightforward, as Equations (2) and (3) can easily be made multivariate functions of the exposures of interest, while any additional exposures are still fixed (or averaged over) as in the univariate case. Extending the ALE approach to two exposures requires a bit more effort. The ALE second-order effect is defined as in Equation (5)

$$f_{k,l, ALE}(z_k, z_l) \equiv h_{k,l, ALE}(z_k, z_l) - \int_{z_{min,k}}^{z_k} \mathbb{E}\left[\frac{\partial h}{\partial Z_k}(Z_k, \mathbf{Z}_l) | Z_k = z'_k\right] dz'_k - \int_{z_{min,l}}^{z_l} \mathbb{E}\left[\frac{\partial h}{\partial Z_l}(Z_k, \mathbf{Z}_l) | Z_l = z'_l\right] dz'_l, \quad (5)$$

where

$$h_{k,l, ALE}(z_k, z_l) \equiv \int_{z_{min,k}}^{z_k} \int_{z_{min,l}}^{z_l} \mathbb{E}\left[\frac{\partial^2 f}{\partial Z_k \partial Z_l}(Z_k, \mathbf{Z}_{[-k-l]}) | Z_k = z'_k, Z_l = z'_l\right] dz'_k dz'_l. \quad (6)$$

Thus, $f_{k,l, ALE}(z_k, z_l)$ excludes the main effects of Z_k and Z_l and solely represents the “second-order” or “interaction” effects. In practice, one might add the main effects for each exposure as defined in Equation (4) back to this quantity to obtain a full picture of the joint effects of the two exposures on the exposure-response function. Once again, all partial derivatives can be approximated with small finite differencing of h or f . For further details regarding the

estimation of the quantities described in this section, we again refer the reader to Apley and Zhu.¹⁹

The benefits of using ALE over the other approaches are threefold: 1) effects of correlated exposures are isolated by targeting partial derivatives of f , 2) estimates are only informed by predictions on plausible exposure profiles since averaging is done using the conditional distribution (i.e., no extrapolation), and 3) the computation is relatively fast compared to PD functions since only two predictions (or six in the case of second-order effects) are needed for each observation, regardless of the number of levels of Z_k being considered.

4 | SIMULATION STUDY

To evaluate the proposed approach, specifically for different BART specifications, we conduct a brief simulation study. We use the populations and locations of the 128 ZIP codes from the first year (2011) of the application. We set confounder effects

$$\beta = (-2, -1, 1, 2)^T \text{ and the true exposure-risk surface function to } f(\mathbf{z}) = -10 + \frac{f_0(\mathbf{z})}{5},$$

where $f_0(\mathbf{z}) = 10 \sin(z_1 z_2) + 20(z_3 - 0.5)^2 + 10z_4 + 5z_5$ is the benchmark function proposed by Friedman.³³ While this surface only depends on five exposures, we generate five additional noise exposures (ten total exposures) to assess the performance in settings where not all exposures are important. Spatial random effects are sampled from their pCAR prior with $\rho = 0.9$ and $\tau^2 = 0.3$. For $i = 1, \dots, 128$ regions and $j = 1, \dots, 300$ observations, we simulate outcomes using the following data generating process:

1. Generate $X_{ij,1}, X_{ij,2}, X_{ij,3}, X_{ij,4} \stackrel{i.i.d.}{\sim} \text{Uniform}(0, 1)$.
2. Generate $\mathbf{Z}_{ij} \sim \text{MVN}(\mathbf{0}_{10 \times 1}, \Sigma_{10 \times 10})$, where $\Sigma_{[1:5, 1:5]}$ matches the observed correlation matrix of PM_{2.5}, NO₂, O₃, CO, and maximum temperature from the application, and all other exposure pairs are independent. Only the first five exposures are used to generate the outcome. All values are scaled to [0, 1] using min-max normalization.
3. Sample $Y_{ij} \sim \text{NB}\left(\xi = 1, p_{ij} = \frac{e^{\eta_{ij}}}{1 + e^{\eta_{ij}}}\right)$ where $\eta_{ij} = \log(\text{Pop}_{ij}) + \mathbf{X}_{ij}^T \beta + f(\mathbf{Z}_{ij}) + \nu_i$

We consider BART ensembles of size $T = \{10, 25, 50, 100\}$, both hard and soft decision rules, and both the classic and sparse branching processes.³⁴ We also consider a parametric alternative to BART, which approximates the functional forms of exposures using natural cubic splines with 4 degrees of freedom for main effects as well as all pairwise linear interaction terms among the exposures. Each setting is repeated 200 times. The average bias, root mean squared error (RMSE), and 95% credible interval coverage for $f(\mathbf{z})$ are presented in Table 1, along with Monte Carlo standard error estimates.

Soft BART had excellent bias, coverage, and RMSE even when few trees were used. When many trees were used ($T = 100$), performance of traditional BART improved, but was still worse than soft BART in terms of coverage and RMSE. In general, increasing the number of trees beyond 25 did not appear to improve the performance of soft BART. Additionally, using the sparse branching process prior mostly resulted in improved coverage

and reduced RMSE (Table 1). This suggests the sparsity-inducing Dirichlet prior was effective at identifying the important exposures and avoiding tree splitting rules based on the noise exposures. The parametric approach (Splines + Linear Interactions) was also competitive, with coverage and RMSE results better than those obtained from the traditional BART ensembles, but not quite as strong as soft BART. Since the true exposure-risk surface function is smooth, this finding is somewhat unsurprising.

We present simulation results for the marginal ALE plots for each exposure using both the parametric approach and the $T = 25$ soft, sparse trees BART setting in Figure 1. On average across simulations, the true functional forms of the five important exposures is recovered remarkably well. The null effects of the noise exposures are also accurately captured by soft BART, primarily due to the ensembles avoiding splitting on these covariates entirely. The parametric approach generally captures the null effects as well, but does not have built-in variable selection. Results for the pairwise ALE plots from the soft BART model are included in Appendix B of the Supplementary Material.

In addition to the excellent performance on the recovery of f , estimates for β, ξ, τ^2, ρ , and ν were also generally unbiased and exhibited reasonable 95% credible interval coverage. Results for these parameters are included in Appendix B of the Supplementary Material.

5 | APPLICATION

We observed 478,311 asthma-related ED visits from 219,136 daily counts during the warm season in Atlanta from 2011-2018. These visits came from 128 ZIP codes from Clayton, DeKalb, Gwinnett, Fulton, and Cobb counties. The number of asthma-related ED visits was relatively stable year-over-year during this time frame, but in general more visits are observed at either end of the warm season (April and October), and occasionally coincide with federal holidays as well. To account for potential confounding by these, we included an indicator variable representing federal holidays and a natural cubic spline on the day-of-year with 7 degrees of freedom per year (one per each warm season month). Given previous findings suggesting the importance of socioeconomic status in this same dataset,⁸ we also include the annual ZIP code-level percent below the poverty threshold as a time-varying linear confounder.

We consider four chemical exposures $\text{PM}_{2.5}$, NO_2 , O_3 , and CO , as well as a meteorological exposure in maximum temperature. Each of these are recorded daily and included as 3-day moving averages. We fit soft BART ensembles of 10, 25, 50, and 100 trees for each of the five primary exposures individually and as a mixture. We run each model for 5,000 burn-in iterations, and then draw 1,000 posterior samples using a thinning interval of 10 iterations (15,000 total MCMC iterations).

For each model, we compute the widely applicable information criterion (WAIC) as an approximation to leave-one-out cross-validation.^{35,36} The results are plotted in Figure 2. For the single-exposure models, the WAIC is lowest for NO_2 and CO , suggesting that these two exposures are the most predictive of asthma-related ED visits when considered individually. As suspected, the mixture model containing all five exposures had a much lower WAIC than

any of the single-exposure models. While increasing the ensemble size beyond 25 trees does not appear to improve the WAIC for any of the single-exposure models, larger ensembles may lead to some improved performance of the mixture model in terms of WAIC. However, when summarizing results of the mixture models with larger ensembles, we found the main findings to be generally similar to the fit with 25 trees. Due to this finding and for the sake of an even comparison, we will consider only the 25-tree single-exposure and mixture models in this section.

A popular approach for assessing the overall mixture effect is to evaluate the exposure-risk surface at a range of exposure values. For instance, one might plot the fitted exposure-risk function while simultaneously setting all exposures to specific quantiles (see Figure 3). Using this strategy, the overall mixture effect suggests a decreasing risk of asthma-related ED visits with increased exposure levels. One challenge with this approach is that it is difficult to assess the contribution of each individual exposure to the overall mixture effect. A larger issue is that these exposure profiles are not particularly realistic. The observed pairwise proportions of exposures across all ZIP code days belonging to the same decile range from just 10% (NO₂ and temperature, CO and temperature) to 23% (NO₂ and CO). Meanwhile, just 0.06% of all ZIP code days had all five exposures in the same decile. This observation underlines the need to evaluate the exposure-risk function in a more realistic manner.

Alternatively, we can avoid extrapolation in our assessment of the mixture effect by referencing the ALE. Estimates of each exposure's ALE shift slightly in the mixture model compared to their single-exposure models (Figure 4). Most notably, the largely null effect of O₃ shifts to harmful in the mixture model. Additionally, in the mixture model, PM_{2.5} has a borderline harmful main effect, NO₂ has a strong negative association with ED visits, and CO and temperature have some upside-down “U-shaped” relationship with ED visits (Figure 4).

In the mixture model, the exposures may interact with one another as well. Here we focus on the potential joint effects of each chemical exposure with temperature, but the resulting fit also showed some interaction among the chemical exposures (see Appendix C of the Supplementary Material). In Figure 5, we note that the estimated ALE for each pollutant depends on temperature to some extent. For instance, the negative association between NO₂ and ED visits is more pronounced at lower temperatures. The positive association between O₃ and ED visits is more pronounced at lower temperatures, unless the O₃ concentration is very high. We also note that the estimated risk associated with PM_{2.5} only appears to differ with temperature for lower PM_{2.5} concentrations, while the reverse is true for CO.

6 | DISCUSSION

In summary, we show through a simulation study and real data application that BART (specifically soft BART) can be used for estimating complex mixture exposure-risk surfaces in the context of count responses, such as visits to an emergency department. Additionally, we have demonstrated the utility of ALE for analyzing marginal and joint effects of 1-2 exposures in the presence of other exposures. Plots such as those included in Figures 4 and 5

are straightforward to interpret individually since the ALE estimation process only averages over plausible exposure profiles supported by the data.

Our application findings regarding asthma-related emergency department visits are also interesting. The exposure concentration required to achieve above average risk may depend on the temperature. We observed interaction between ozone and temperature - specifically a stronger ozone effect at cooler warm-season temperatures. While the analysis framework we have proposed is not causal in nature, we hypothesize that modification of chemical exposure effects by temperature could be related to individual-level behavior - e.g., people may be less likely to experience the effects of air pollution on very hot days where they are more inclined to stay indoors. We also found a strong negative association between NO₂ and ED visits, which stands in contrast to some findings regarding NO₂ and respiratory outcomes. Due to the dense tree canopy and high traffic emissions in Atlanta, NO₂ and volatile organic compounds are a precursor to ozone, and higher NO₂ levels may be reflective of warmer days with lower ozone pollution.

One current limitation of our methodology is the lack of ability to formally detect lagged effects. Since in our application we are focused on short-term effects, using 3-day moving averages for the daily exposures is sufficient. In general it is difficult to simultaneously estimate non-linear, interaction, and lagged effects in mixture modeling. Regardless of the specific model selected, we find it important to consider exposures as a mixture rather than individually both when it comes to estimating the exposure-response surface and interpreting the resulting fit.

Our modeling approach is also computationally demanding. Drawing Pólya-gamma latent variables for every observation in the data augmentation step and updating the soft BART ensemble takes time. The model shared in the application took approximately 20 hours to fit and summarize. While this is a long time, Gaussian process based modeling approaches (e.g. BKMR) would be infeasible given the sample size of 220,000. A BART ensemble using traditional “rigid” trees would also fit faster than soft BART, but may also require a greater number of trees to achieve comparable performance.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

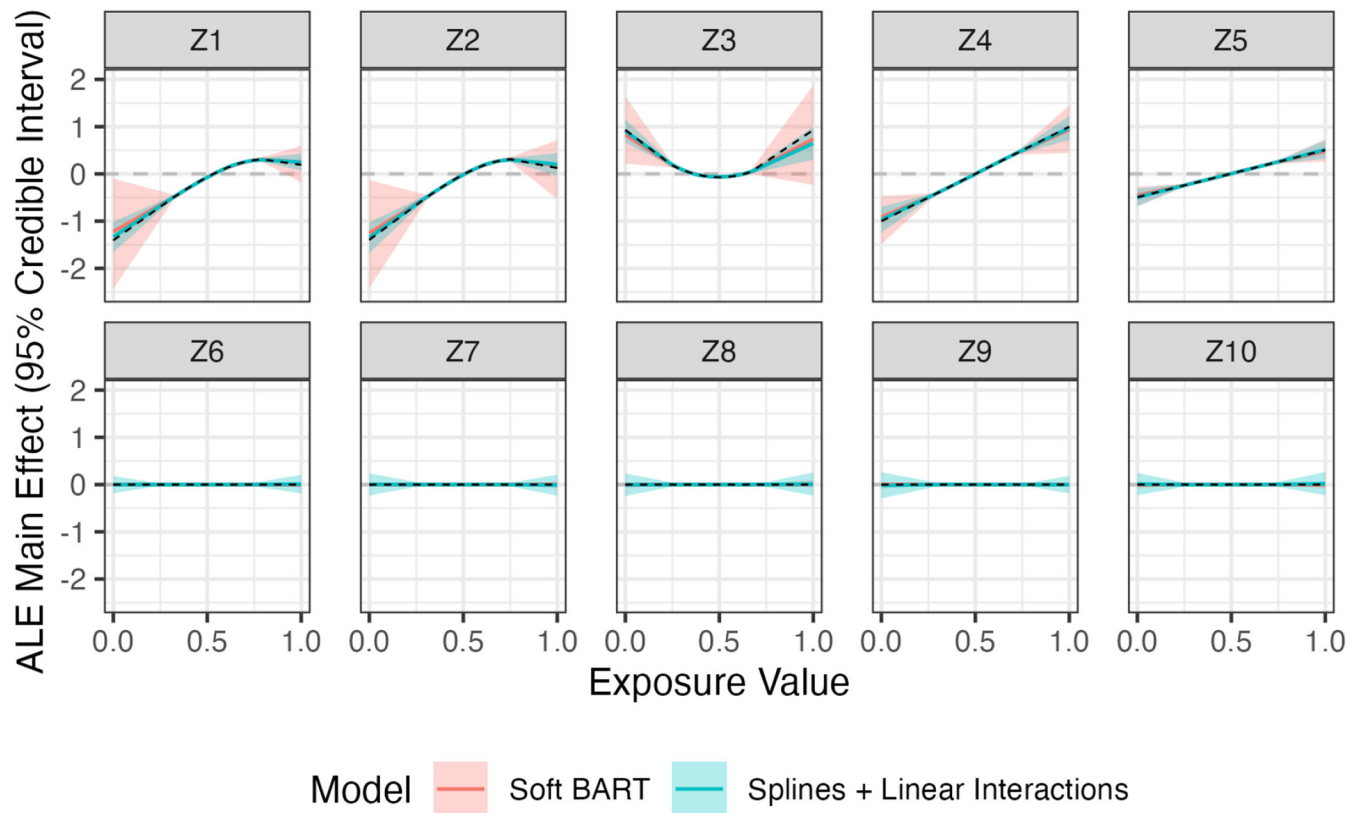
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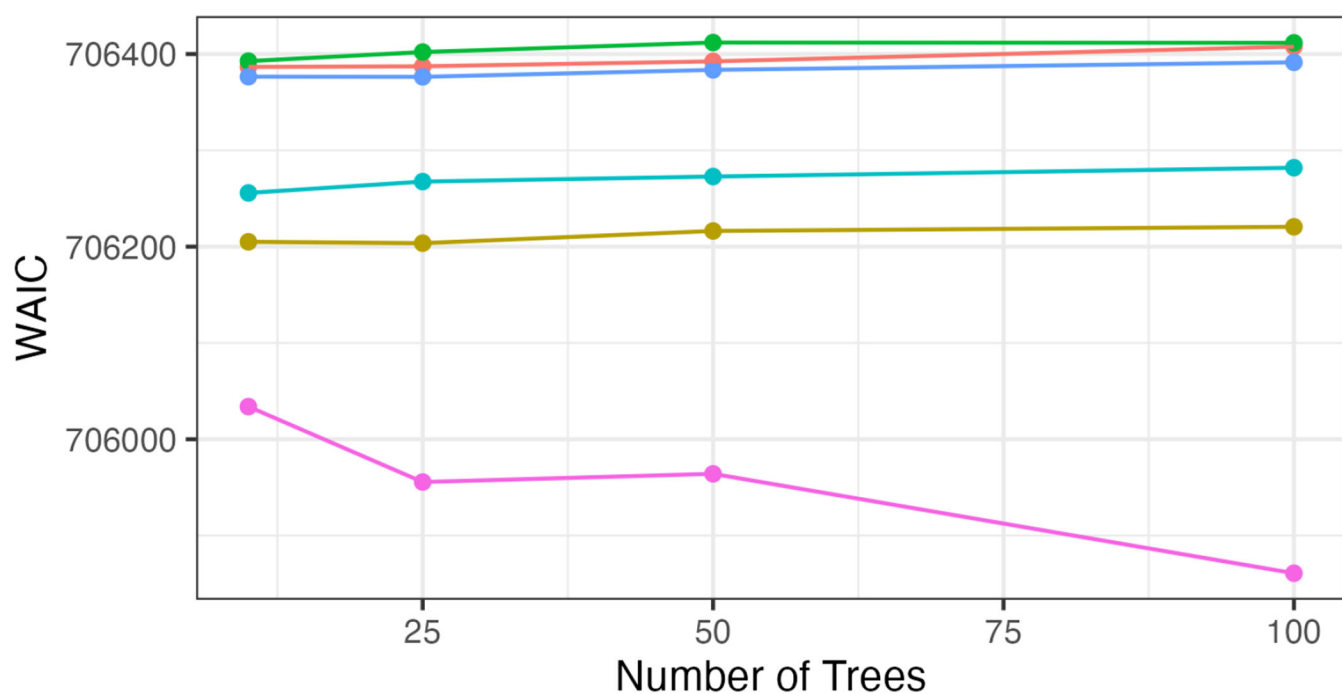
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**FIGURE 1.**

ALE main effects from the simulation study using BART ($T = 25$ soft, sparse trees) in red, and a parametric regression (natural cubic splines for each exposure and all pairwise linear interactions) in blue. Solid lines represent the average ALE point estimate, and ribbons represents the average 95% posterior credible interval bounds across 200 simulations. The true ALE is depicted as a dashed black line. ALEs are calculated using $K = 40$ quantile intervals for each exposure.



Exposures

- PM25
- O3
- Tmax
- NO2
- CO
- PM25 + NO2 + O3 + CO + Tmax

FIGURE 2. Widely applicable information criterion (WAIC)^{35,36} for the single-exposure and mixture models fit to the Atlanta ED visit data.

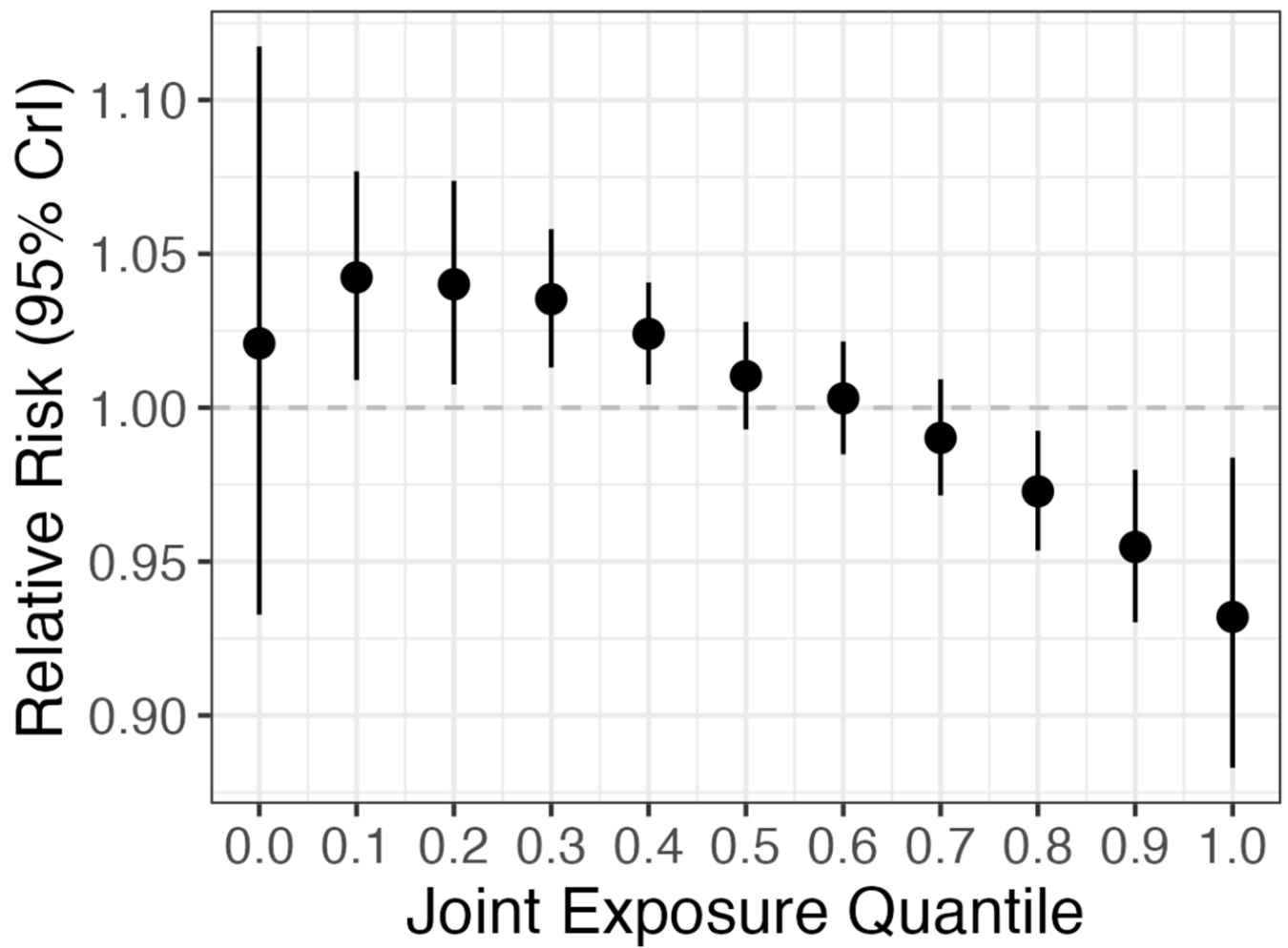
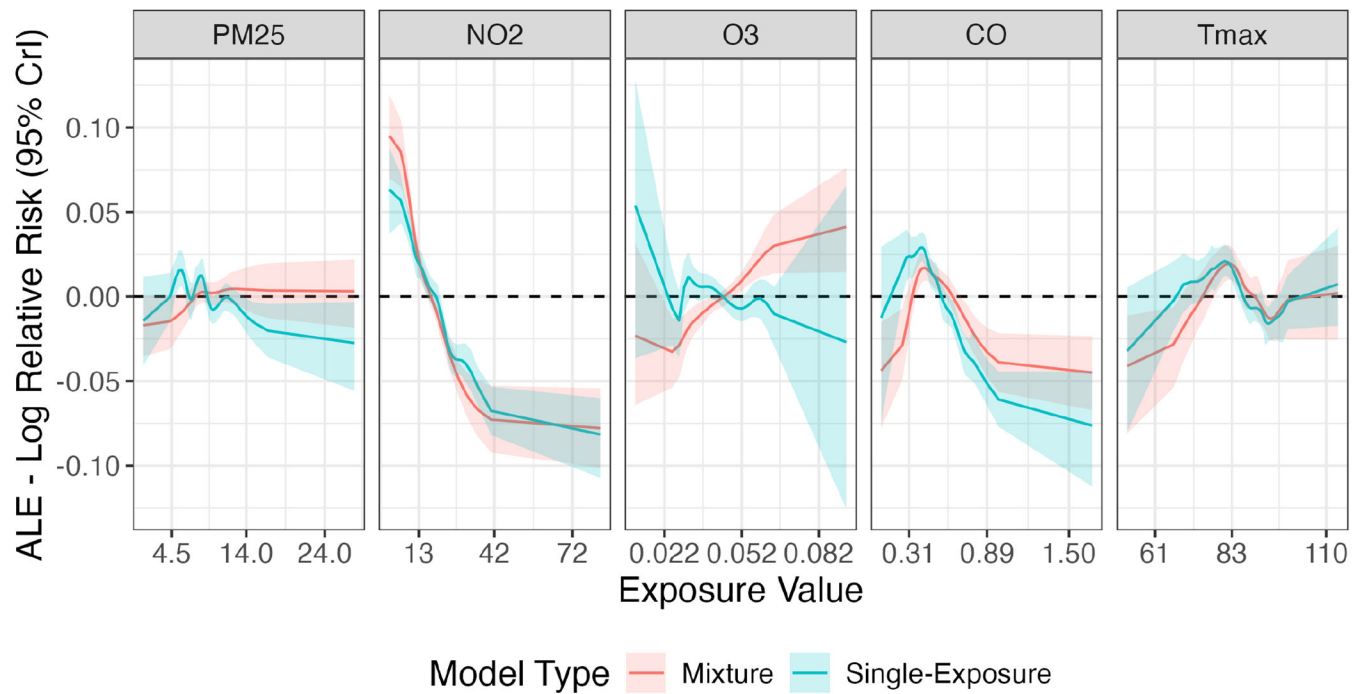
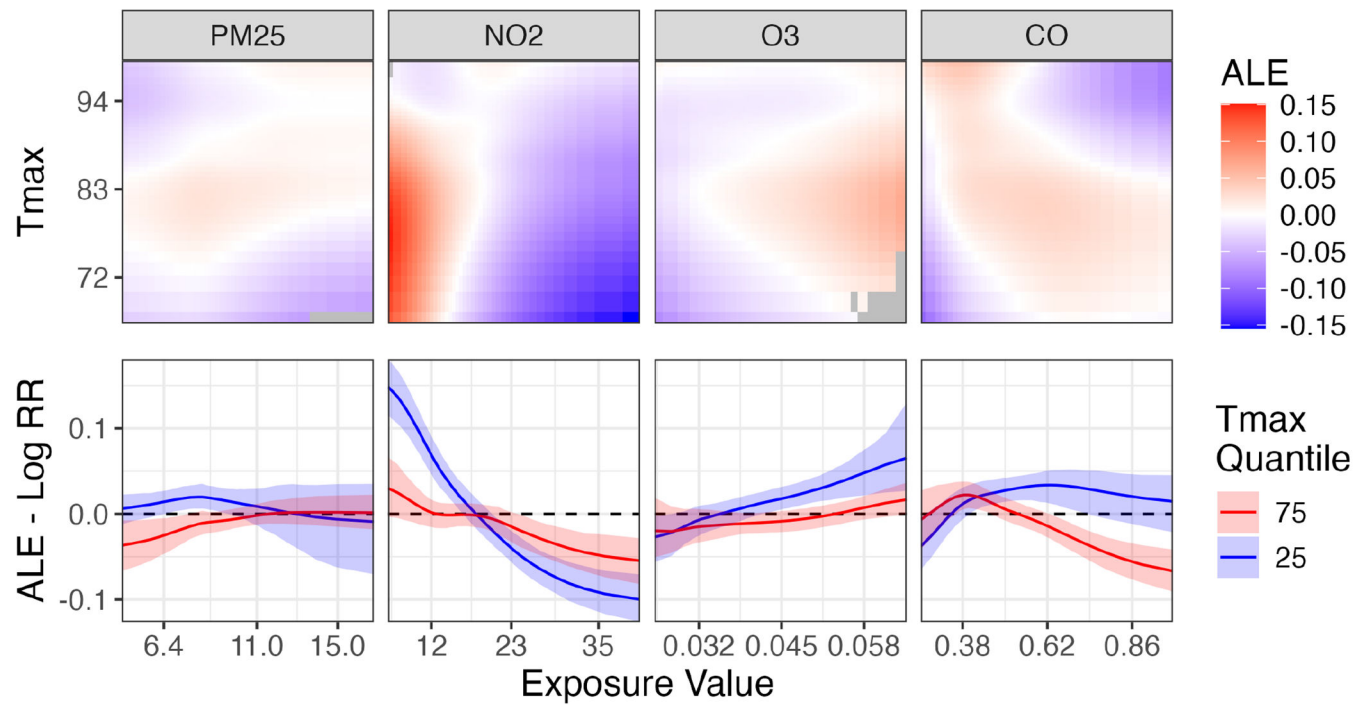


FIGURE 3.

Estimated relative risk (and 95% posterior credible interval) of an asthma-related ED visit when all exposures are simultaneously set to the same decile. Estimates are relative to the exposure profiles corresponding to the overall average risk.

**FIGURE 4.**

Main effect ALE for exposures individually (blue) and as part of a mixture (red) in the Atlanta application. Plots are centered so that zero on the y-axis represents an average risk level. ALEs are calculated using $K = 40$ quantile intervals for each exposure.

**FIGURE 5.**

Pairwise second-order ALE for each chemical exposure with maximum temperature for the mixture model with $T=25$, with the corresponding main effect ALEs added on. The top row includes the posterior mean ALE for all observed pairwise combinations, while the bottom row plots horizontal slices at the first and third quartiles of maximum temperature along with 95% credible intervals. ALEs are calculated using $K=40$ quantile intervals for each exposure. Plots are trimmed so that only the central 95% of each exposure is displayed.

TABLE 1
Average bias, average 95% CrI coverage, and RMSE for $f(\mathbf{Z})$ predictions across simulations.

Model Type	$T^{\dagger}$	Soft [‡]	Sparse [§]	Bias ×10 (MCSE)	Coverage (MCSE)	RMSE ×10 (MCSE)
Splines + Linear Interactions	—	—	—	−0.02 (0.0135)	0.90 (0.0209)	0.67 (0.0033)
BART	10			0.11 (0.0138)	0.88 (0.0229)	1.17 (0.0050)
BART	10		✓	0.11 (0.0142)	0.96 (0.0143)	1.06 (0.0043)
BART	10	✓		0.02 (0.0139)	0.91 (0.0207)	0.41 (0.0065)
BART	10	✓	✓	0.02 (0.0137)	0.92 (0.0188)	0.41 (0.0061)
BART	25			0.05 (0.0141)	0.78 (0.0293)	0.98 (0.0037)
BART	25		✓	0.05 (0.0138)	0.89 (0.0221)	0.89 (0.0039)
BART	25	✓		0.01 (0.0139)	0.94 (0.0167)	0.39 (0.0056)
BART	25	✓	✓	0.01 (0.0137)	0.95 (0.0155)	0.37 (0.0056)
BART	50			0.02 (0.0136)	0.81 (0.0279)	0.89 (0.0035)
BART	50		✓	0.02 (0.0139)	0.88 (0.0228)	0.82 (0.0037)
BART	50	✓		0.01 (0.0137)	0.95 (0.0147)	0.42 (0.0050)
BART	50	✓	✓	0.02 (0.0140)	0.96 (0.0140)	0.37 (0.0054)
BART	100			0.01 (0.0140)	0.88 (0.0233)	0.85 (0.0035)
BART	100		✓	0.01 (0.0137)	0.92 (0.0188)	0.77 (0.0037)
BART	100	✓		0.02 (0.0138)	0.96 (0.0131)	0.46 (0.0047)
BART	100	✓	✓	0.01 (0.0136)	0.96 (0.0130)	0.39 (0.0054)

CrI: Bayesian posterior credible interval.
MCSE: Monte Carlo Standard Error.
* Number of trees.
[†]Soft BART used¹⁷.
[‡]Sparse branching process used³⁴.